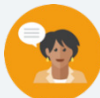


13.8 Identifying and Interpreting Parts of an Expression or Equation – Part 1

Lesson Introduction

 Benchmark	MA.912.AR.1.1 Identify and interpret parts of an equation or expression that represent a quantity in terms of a mathematical or real-world context, including viewing one or more of its parts as a single entity.		 Learning Targets	- I can identify parts of an expression, including factors, terms, constants, coefficients, and variables. - I can interpret parts of an equation or expression that represent a quantity in terms of a mathematical or real-world context.
 Benchmark Coherence	Building On			Working Toward
	N/A			N/A
 Essential Questions	- How can I determine if a part of an expression is a factor, term, constant, coefficient, or variable? - How do I interpret parts of an equation or expression that represent a quantity in terms of a mathematical or real-world context?			
 Lesson Narrative	This lesson is the first part of a two-part lesson series. In this lesson, students learn about the parts of an equation and expression, including equations, expressions, functions, constants, coefficients, terms, initial value, rate of change, and quantity. The instruction during this lesson is primarily centered around real-world problems where students will extract information from.			
 Component Summary	13.8.1 Warm-Up (~5 min.)	Florida scrub-jay data activity (<i>SE p. 279</i>)	13.8.3 Guided Practice (10-15 min.)	Parts of an expression practice activity (<i>SE p. 281</i>)
	13.8.2 Guided Instruction (15-20 min.)	Introduction to parts of an equation or expression (<i>SE p. 279</i>)	13.8.4 Your Turn! (10-15 min.)	Independent practice with parts of an equation (<i>SE p. 282</i>)
	13.8.5 Wrap-Up (~5 min.)	Self-reflection activity (<i>SE p. 282</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		N/A	This activity is the first part of a two-part lesson series. Review both lessons prior to teaching either to know where one starts and stops in comparison to the overall learning objectives.

 Teacher Tips	Common Misconceptions - Students may reverse the definition of an expression and an equation. - Students may mistake a constant for a coefficient and vice versa. - Students may misunderstand the concept of “term” when there is only one term. - Students may reverse the initial value and rate of change values.	Things to Consider This lesson is comprised of real-world word problems. This can often be an initial barrier for students. Consider leveraging word-problem strategies during this lesson to help students overcome this component.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Three Reads This lesson consists completely of word problems. Consider the use of the Three Reads routine to help students navigate these types of problems. This routine supports reading comprehension of problems and meta-awareness of mathematical language.	MA.K12.MTR.1.1: Participate in Effortful Learning This lesson consists of complex word problems that may cause students to shut down or become overwhelmed with the challenge. During this lesson, students will build perseverance by modifying methods as needed while solving a challenging task. Students should stay engaged and maintain a positive mindset when working to solve tasks.
 Differentiate Instruction	Students who are struggling readers may find this lesson to be a challenge due to the amount of text and reading comprehension necessary. Consider differentiating the problems using reading and word-problem strategies that are most appropriate for your students and their needs. This lesson provides several opportunities for spiraled review. Consider student performance in past units when working through this lesson.	
Lesson Notes:		

13.8.1 Warm-Up: Florida Scrub-Jay

Component Overview: Students will review provided information about the Florida scrub-jay and answer three questions using the provided data.



13.8.1 Warm-Up: Florida Scrub-Jay

The Florida scrub-jay is a blue and gray bird about the size of a blue jay. The Florida scrub-jay is not only native to Florida, but it only lives in Florida. Scrub-jays inhabit the ancient sandy ridges that run down the middle of the state, old sand dunes along Florida's coasts, and sandy deposits along the rivers in the interior of the state. In the absence of lightning-ignited, low-intensity fires that used to be common in Florida, the scrub-jay's natural habitat has become too dense and tall to support them. Before Florida was settled in 1800, the bird's population is estimated to have been 54,000. It's estimated that by 1890 the population was 27,600. More recently, the estimates of the Florida scrub-jay population were 4,000 in 1993 and 3,000 in 2010.



Consider leveraging a Think-Pair-Share routine with this activity. Allow students a couple minutes to try this activity on their own and then, when time is called, work with a partner to review answers and their strategies for approaching the patterns. Allow an opportunity for students to work with other pairs or have a whole-group discussion to discuss strategies and approaches.

1. What is the scrub-jay's population decline from 1800 to 1890 as a percentage?
48.9%
2. What is the rate of decay from 1993 to 2010?
25%
3. During which time period did the most devastating decrease in the Florida scrub-jay population occur?
Answers will vary. Students may say between 1800-1890 because there is a greater overall decline over that time period, but the per year decline is actually greater between 1993-2010.

13.8.2 Guided Instruction: Parts of an Equation or Expression

Component Overview: Students will learn about parts of an equation or expression using four unique, multipart problems that scaffold this concept using both equations and expressions and multiple components.



13.8.2 Guided Instruction: Parts of an Equation or Expression

1. Jacquie is hiking a section of the Florida Trail, starting at the trailhead located at historic Fort Pickens in Gulf Islands National Seashore. Jacquie packs trail mix and beef jerky for snacks. The trail mix provides 3.25 calories per gram and the beef jerky provides 2.7 calories per gram. She would like to only consume 158 calories.

a. $3.25m + 2.7b$ is a(n)

- ☐ equation.
- ☒ expression.
- ☐ function.

b. The values 3.25 and 2.7 are

- ☐ constants.
- ☒ coefficients.
- ☐ terms.

c. The monomials $2.7b$ and $3.25m$ are

- ☐ constants.
- ☐ coefficients.
- ☒ terms.

d. $3.25m + 2.7b = 158$ is a(n)

- ☒ equation.
- ☐ expression.
- ☐ function.

- e. Determine the amount of trail mix Jacquie can eat if she eats 20 grams of beef jerky. Show your work.

$$3.25m + 2.7b = 158$$

$$3.25m + 2.7(20) = 158$$

$$3.25m + 54 = 158$$

$$3.25m + 54 - 54 = 158 - 54$$

$$3.25m = 104$$

$$\frac{3.25m}{3.25} = \frac{104}{3.25}$$

$$m = 32$$

$$m = 32$$

Jacquie can eat 32 grams of trail mix.



What is the difference between an equation and an expression?



Consider the use of a word wall or unit glossary. These terms are not new, but having a reference guide available, even if a word wall or anchor chart, can provide students a visual tool to use as they progress through this lesson series.

13.8.2 Guided Instruction: Parts of an Equation or Expression (continued)

2. U.S. food availability is determined by taking how much of a certain food is produced and subtracting exports to other countries. In 1970, the amount of cheese available for consumption was 17.22 pounds per person. For each year after 1970, the amount of cheese that was available increased by 0.48 pounds per person.

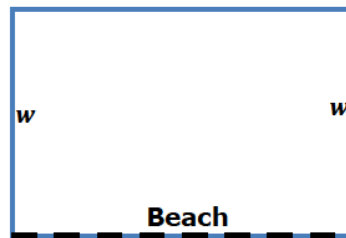
- a. This situation represents
relationship.

- ☐ a linear
- ☐ a quadratic
- ☐ an exponential

- b. Complete the table.

Quantity	Description
0.48	☐ initial value ☐ rate of change ☐ informational only
17.22	☐ initial value ☐ rate of change ☐ informational only
1970	☐ initial value ☐ rate of change ☐ informational only

3. A lifeguard uses 2,000 feet of rope to create a safe rectangular-shaped swimming area in the ocean. The diagram shows the width, w , of the swimming area and the location of the beach. The rope is not used alongside the beach.



Question 2 is a review of information from prior units. This provides an opportunity to determine which of the relationships students do not understand if they get the answer incorrect. Question 2b assesses students' abilities to interpret the key features of the function.



Consider releasing question 3 for students to work with a partner to try this one on their own. This can serve as an opportunity to provide productive struggle and collaborative problem-solving opportunities.

13.8.2 Guided Instruction: Parts of an Equation or Expression (continued)

Explain what each quantity represents in this context.

Quantity	Explanation
$2000 - 2w$	<i>The initial amount of rope was 2000 is subtracted by the widths of the rectangle represented by $2w$. This represents the length of the swimming area.</i>
$w(2000 - 2w)$	<i>This represents the Area formula. "w" at the front represents the width.</i>

4. Microplastics are buoyant plastic materials smaller than 0.5 centimeters in diameter. The amount of these plastics in the ocean since 1994 can be modeled by the equation $y = 100,100(1 + 0.073)^x$, where y is the number of tons of microplastics in the ocean and x the number of years since 1994.

- a. Discuss with your partner why the values 0.5 and 1994 are not in the equation $y = 100,100(1 + 0.073)^x$. Complete the table.

Value	Explanation
0.5	<i>This is just telling the size of the plastic materials.</i>
1994	<i>This is just information to tell the year that they began the model. This is used in calculations when solving the problem, not setting it up.</i>

- b. Ask a classmate for their explanation. Record their explanation. *You are the only person who should write in the second column of the table.* Have the person initial next to your summary stating that your summary is correct.

Value	My Summary of Their Explanation	Initials
0.5	<i>Answers will vary.</i>	
1994	<i>Answers will vary.</i>	



Watch for student misconceptions with question 4. Students often take any number and use it when doing word problems. This question is to help be sure students understand that not every number in a problem is used in the expression, equation, or function.



For question b, prompt students to compare their explanations with students they have not worked with yet in this unit to allow for additional viewpoints.

13.8.2 Guided Instruction: Parts of an Equation or Expression (continued)

c. Complete the sentences.

The equation $y = 100,100(1 + 0.073)^x$ represents

- ☐ exponential decay.
- ☒ exponential growth.

The rate of ☐ decay
☒ growth is 7.3%.

The value 100,100 is the ☐ amount of plastics
☒ tons of microplastics

found in the ocean in 1994.



Question c is a spiraled refresher to exponential functions. Students should be able to refer back to their notes with exponential functions if additional review is necessary.

13.8.3 Guided Practice: Parts of an Equation or Expression

Component Overview: Students will complete two practice problems associated with parts of an equation or expression.



13.8.3 Guided Practice: Parts of an Equation or Expression

1. An orange grove manager uses two different-sized buckets to load oranges into a truck. The first bucket can transport x pounds of oranges, and the second bucket can transport y pounds of oranges. The first bucket is filled up G times and the second is filled up H times.

Explain what each quantity means.

Quantity	Explanation
$G + H$	G – the number of times the first bucket is filled. H – the number of times the second bucket is filled. $G + H$ represents the total amount of times the first and second buckets are filled.
$x + y$	x – pounds of oranges in the first bucket y – pounds of oranges in the second bucket $x + y$ represents the total amount of pounds of oranges in both buckets.
$Gx + Hy$	The total number of pounds of oranges that are loaded onto the truck.



Watch for misconceptions in the explanations. Push for specific language and not general language, as students can often use general language to hide that they aren't confident in their answers or are unsure of the actual answer.

2. A pattern is shown. The number of tiles in each step is represented by the expression $(n + 1)^2 + 3$, where n is the step.

Step 1	Step 2	Step 3

Circle the part of each diagram that is represented by the expression $(n + 1)^2$.



Students may struggle with the balance between the provided expression and the visual representation.

13.8.4 Your Turn!

Component Overview: Students will use a real-world scenario to complete four questions.



13.8.4 Your Turn!

The function $C(x) = 0.48x + 17.22$ models the amount of cheese, C , in pounds, available for consumption for each year since 1970.

1. What does the slope of the function, 0.48, represent in this context?
There was an increase of 0.48 pounds of cheese consumption each year.
2. Which value is a constant?
17.22
3. What does the constant represent?
The amount of cheese consumed in the year 1970.
4. What does the variable x represent?
The number of years since 1970.



Consider using the time in this activity to work with a small group of students who performed poorly during the 13.8.3 Guided Practice activity.



When reviewing the answers to this activity, have students share out how they knew their answer was correct. Students who are struggling can often learn from the approaches of their peers.

13.8.5 Wrap-Up: Reflection

Component Overview: Students will complete a three-part reflection activity regarding this lesson.



13.8.5 Wrap-Up: Reflection

1. Complete the table.
Answers will vary.

Today I deserve  for ...	
Today I  by doing ...	
Today I was  by ...	



Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.